

CHUN-KAI HUANG (JEFF)

Curriculum Vitae

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EDUCATION

National Taiwan University (NTU), Taipei, Taiwan

Sep 2008–Jun 2014

M.S. in Mechanical Engineering (ME), System and Control Division, 2014

- Overall GPA: 4.28/4.3, Major GPA: 4.24/4.3, Rank: 1/33 in the System and Control Division

B.S. in Mechanical Engineering (ME), 2012

- Overall GPA: 4.11/4.3, Major GPA: 4.14/4.3, Rank: 4/153 in the graduating class

Stanford University, California, USA

Aug 2012

American Language and Culture Program

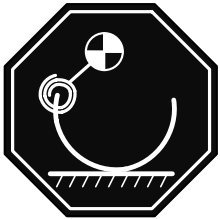
RESEARCH EXPERIENCE

Research Areas

- Robotics and bio-inspired robots
- Mathematical legged models and model-based control strategy
- System control and stability analysis

Thesis Development of Dynamic Legged Models with Rolling Contact and Their Role as Templates for Inducing Dynamic Gaits on a Hexapod Robot

- Advisor: Dr. Pei-Chun Lin (Bio-inspired Robotics Laboratory)



Reduced-order Models (Main work in MS study), MST, Taiwan

2012–2014

- Analyzed the stability property of the Rolling SLIP (R-SLIP) model, which is a 2-DoF dynamic model with compliance and rolling contact
- Developed the Two-rolling-leg (TRL) model, which is a 3-DoF dynamic model with rigid body and two compliant rolling legs
- Investigated the gait-level stability property of the TRL model



Bio-inspired Legged Robots (Main work in MS study), MST, Taiwan

2012–2014

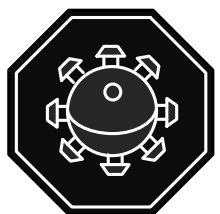
- Revealed the mapping between dynamic models and the experimental robot
- Designed a stability-based and model-based control strategy to excite dynamic locomotion of the empirical robot, including jogging, pronking and bounding
- Introduced a model-based state estimator, which relies on IMU and encoder feedback, as a feedback strategy to improve the stability of the robot's motion



Chicken Head Project, Introduction to Robotics, Dept. of ME, NTU

2012

- Developed the control strategy, based on MATLAB codes, to keep track of a point by utilizing sensory feedback, such as camera and IMU
- Designed a 2-DoF robot arm; Skills included laser cutting, mechanical design, signal processing and building the mechatronic system for driving 2 DC motors
- Programmed the LabVIEW codes to govern the motion of the robot arm, including inverse kinematics and trajectory planning



Spherical Robot, College Student Participation in Research Projects, MST

2011–2012

- Constructed an omnidirectional spherical robot; Skills including laser cutting, glass fiber processing, lost foam casting and operating the winding machine
- Built the control strategy and mechatronic system to actuate 12 electromagnets by commanding different PWM signals to each electromagnet
- Programmed the LabVIEW codes to control the motion of the robot

HONORS AND AWARDS

- 1st Place, 2014 Master's Thesis Award, Robotics Society of Taiwan (RST) 2014
- Best Paper Award, MathWorks 2014 MATLAB & Simulink Tech Forum & Expo, Taiwan 2014
- 1st Place, MathWorks 2014 MATLAB & Simulink Paper Contest, TeraSoft Inc. 2014
- Presentation, 2013 IEEE Int. Conf. on Intelligent Robotics and Systems (IROS), Tokyo, Japan 2013
- Assistantship, Ministry of Science and Technology (MST), Taiwan 2012–2014
- College Student Research Creativity Award, MST, Taiwan 2012
- Undergraduate Research Assistantship, MST, Taiwan 2011–2012
- Scholarship (Prize: 3333 USD, highest in Taiwan), Lin-Hsiung-Cheng Foundation, Taiwan 2011
- Five times Academic Excellence Award (top 5% of the class), NTU 2009–2012

PUBLICATIONS

Journal Articles

- K. J. Huang¹, C. K. Huang¹, and P. C. Lin, "A simple running model with rolling contact and its role as a template for dynamic locomotion on a hexapod robot," *Bioinspiration and Biomimetics*, vol. 9, p. 046004, 2014. (¹These two authors contributed equally)
- C. K. Huang, C. J. Hu and P. C. Lin, "Gait-level Behavioral Analysis of a Sagittal Plane Model with Rolling Contact and its Role as a Template for Bounding and Pronking on a Quadruped Robot," *The International Journal of Robotics Research*. (In review)
- C. J. Hu, C. K. Huang, and P. C. Lin, "TDR-SLIP: a Torque-Actuated Dissipative Spring Loaded Inverted Pendulum Model with Rolling Contact," *Bioinspiration and Biomimetics*. (In preparation)

Conference Proceedings

- C. K. Huang, K. J. Huang, and P. C. Lin, "Rolling SLIP model based running on a hexapod robot," in *IEEE International Conference on Intelligent Robotics and Systems (IROS)*. Tokyo, Japan, 2013, pp. 5608-5614.

TEACHING EXPERIENCE

Full-time Teaching Assistant, Dept. of ME, NTU

Aug 2014–Present

- Instructed the course Measurement and Mechanical Engineering Laboratory
- Exp. of Op-Amp circuits: inverting circuits, instrumentation amplifier, charge amplifier, low pass filter
- Exp. of load cells: developed the strain gage transducer by using Wheatstone bridge circuits
- Exp. of piezoelectric sensors: investigated PVDF strain property and built PZT accelerometer
- Exp. of piezoresistive sensors: examined the characteristics of the piezoresistive pressure sensor

Teaching Assistant, Introduction to Robotics, Dept. of ME, NTU

Sep 2013–Jan 2014

- Prepared course materials on topics including inverse kinematics, Jacobians, manipulator dynamics and trajectory planning algorithm

Teaching Assistant, Conservation Service Learning, Extracurricular Activities Section, NTU

2014 winter

- Led student volunteers to build and maintain nature trails

Teaching Assistant (3 semesters), Badminton-basic, Dept. of Athletics, NTU

Feb 2013–Jun 2014

- Coached the badminton classes

SKILLS

Product Prototyping

- Machining, laser cutting, 3D printing, electrical discharge machining, winding machine

Programming Languages and Software

- MATLAB, LabVIEW, DREAMWEAVER*, SolidWorks, AutoCAD, Visual C++, Ansoft Maxwell

* Designed and maintained the lab website: <http://biorola.me.ntu.edu.tw/>

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Initiator (1st President and 2nd Vice President), Nature Trail Club, NTU

Sep 2013–Jun 2014

- Established the first student association engaging in natural trail conservation in Taiwan

Webmaster, Bio-inspired Robotics Laboratory, NTU

Feb 2013–Jan 2014